

Tutorial 6

Rotational Dynamics : Rolling; Heavy Pulley-Mass System

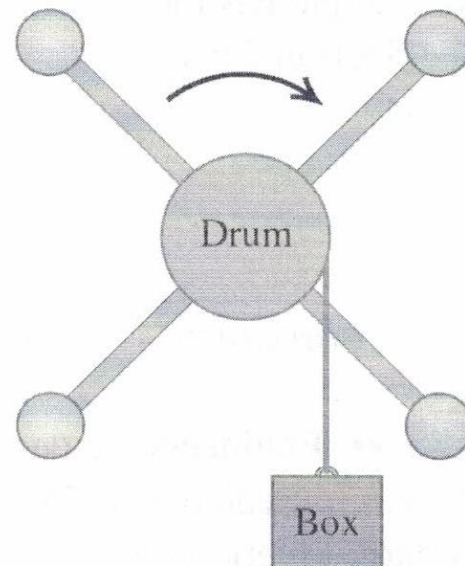
Note:

You may assume the rotation or rolling is without slipping or sliding unless indicated by the question otherwise.

Q9.6 A flywheel rotates with constant angular velocity. Does a point on its rim have a tangential acceleration? A radial acceleration? Are these accelerations constant in magnitude? In direction? In each case give the reasoning behind your answer.

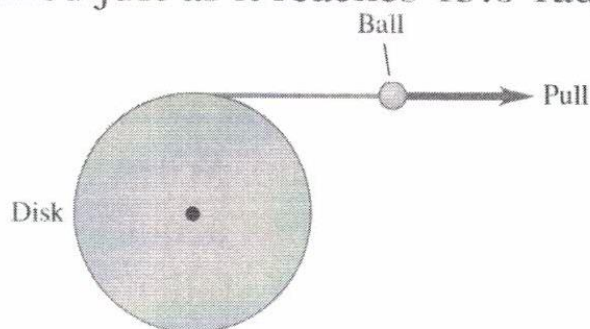
Q9.18 An elaborate pulley consists of four identical balls at the ends of spokes extending out from a rotating drum (Fig. Q9.18). A box is connected to a light thin rope wound around the rim of the drum. When it is released from rest, the box acquires a speed V after having fallen a distance d . Now the four balls are moved inward closer to the drum, and the box is again released from rest. After it has fallen a distance d , will its speed be equal to V , greater than V , or less than V ? Show or explain why.

Figure **Q9.18**



Q9.21 A wheel is rotating about an axis perpendicular to the plane of the wheel and passing through the center of the wheel. The angular speed of the wheel is increasing at a constant rate. Point A is on the rim of the wheel and point B is midway between the rim and center of the wheel. For each of the following quantities, is its magnitude larger at point A or at point B , or is it the same at both points? (a) angular speed; (b) tangential speed; (c) angular acceleration; (d) tangential acceleration; (e) radial acceleration. Justify each of your answers.

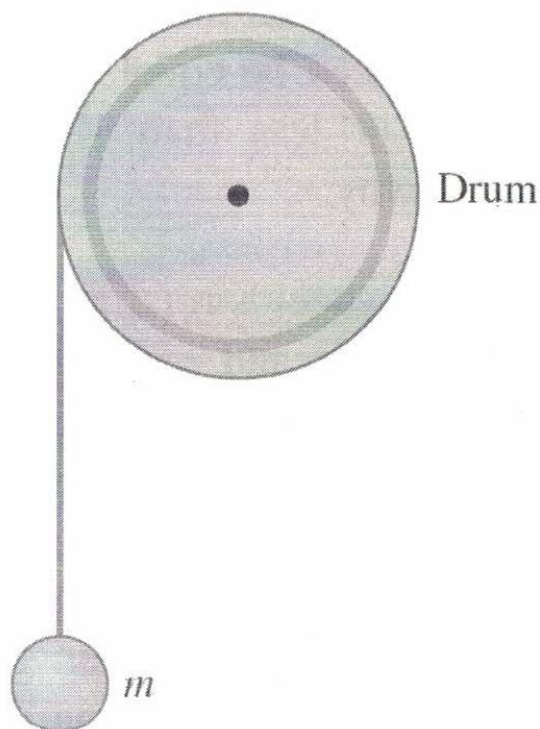
9.65 •• CP CALC A disk of radius 25.0 cm is free to turn about an axle perpendicular to it through its center. It has very thin but strong string wrapped around its rim, and the string is attached to a ball that is pulled tangentially away from the rim of the disk (Fig. P9.65). The pull increases in magnitude and produces an acceleration of the ball that obeys the equation $a(t) = At$, where t is in seconds and A is a constant. The cylinder starts from rest, and at the end of the third second, the ball's acceleration is 1.80 m/s^2 . (a) Find A . (b) Express the angular acceleration of the disk as a function of time. (c) How much time after the disk has begun to turn does it reach an angular speed of 15.0 rad/s ? (d) Through what angle has the disk turned just as it reaches 15.0 rad/s ? (*Hint:* See Section 2.6.)



Ans: (a) 0.600 m/s^3 (b) $\alpha = (2.40 \text{ rad/s}^3)t$ (c) 3.54 s (d) 17.7 rad

9.68 •• Engineers are designing a system by which a falling mass m imparts kinetic energy to a rotating uniform drum to which it is attached by thin, very light wire wrapped around the rim of the drum (Fig. P9.68). There is no appreciable friction in the axle of the drum, and everything starts from rest. This system is being tested on earth, but it is to be used on Mars, where the acceleration due to gravity is 3.71 m/s^2 . In the earth tests, when m is set to 15.0 kg

Figure **P9.68**



and allowed to fall through 5.00 m, it gives 250.0 J of kinetic energy to the drum. (a) If the system is operated on Mars, through what distance would the 15.0-kg mass have to fall to give the same amount of kinetic energy to the drum? (b) How fast would the 15.0-kg mass be moving on Mars just as the drum gained 250.0 J of kinetic energy?

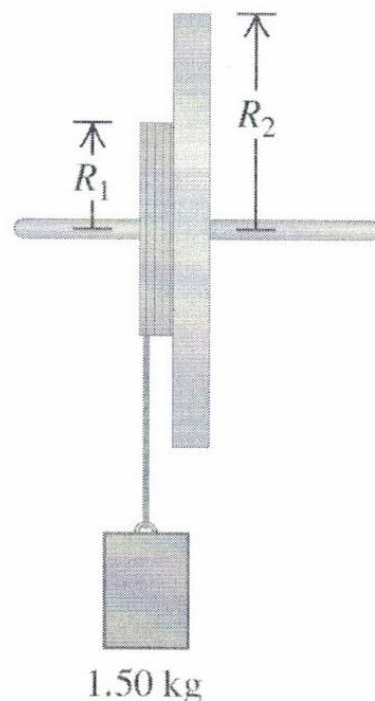
Ans: (a) 13.2 m (b) 8.04 m/s

9.74 •• A sphere consists of a solid wooden ball of uniform density 800 kg/m^3 and radius 0.30 m and is covered with a thin coating of lead foil with area density 20 kg/m^2 . Calculate the moment of inertia of this sphere about an axis passing through its center.

Ans: 4.61 kg m^2

9.87 •• Two metal disks, one with radius $R_1 = 2.50 \text{ cm}$ and mass $M_1 = 0.80 \text{ kg}$ and the other with radius $R_2 = 5.00 \text{ cm}$ and mass $M_2 = 1.60 \text{ kg}$, are welded together and mounted on a frictionless axis through their common center (Fig. P9.87). (a) What is the total moment of inertia of the two disks?

(b) A light string is wrapped around the edge of the smaller disk, and a 1.50-kg block is suspended from the free end of the string. If the block is released from rest at a distance of 2.00 m above the floor, what is its speed just before it strikes the floor? (c) Repeat the calculation of part (b), this time with the string wrapped around the edge of the larger disk. In which case is the final speed of the block greater? Explain why this is so.



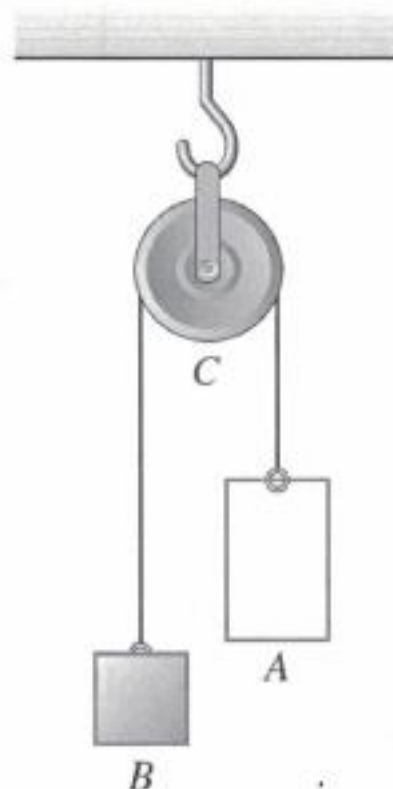
Ans: (a) $2.25 \times 10^{-3} \text{ kg m}^2$ (b) 3.40 m/s (c) 4.95 m/s

Q10.2 Can a single force applied to a body change both its translational and rotational motion? Explain.

Q10.18 A hoop, a uniform solid cylinder, a spherical shell, and a uniform solid sphere are released from rest at the top of an incline. What is the order in which they arrive at the bottom of the incline? Does it matter whether or not the masses and radii of the objects are all the same? Explain.

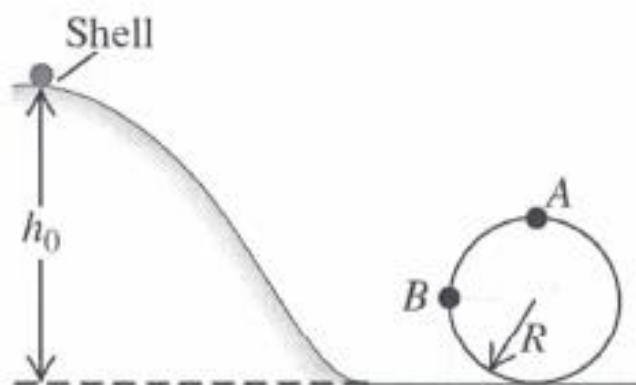
10.67 •• Atwood's Machine. Figure P10.67 illustrates an Atwood's machine. Find the linear accelerations of blocks A and B , the angular acceleration of the wheel C , and the tension in each side of the cord if there is no slipping between the cord and the surface of the wheel. Let the masses of blocks A and B be 4.00 kg and 2.00 kg , respectively, the moment of inertia of the wheel about its axis be $0.300\text{ kg}\cdot\text{m}^2$, and the radius of the wheel be 0.120 m .

Ans: $a = 0.730\text{ m/s}^2$, $T_A = 36.3\text{ N}$,
 $\alpha = 6.08\text{ rad/s}^2$, $T_B = 21.1\text{ N}$



10.76 •• CP A thin-walled, hollow spherical shell of mass m and radius r starts from rest and rolls without slipping down the track shown in Fig. P10.76. Points A and B are on a circular part of the track having radius R . The diameter of the shell is very small compared to h_0 and R , and the work done by rolling friction is negligible. (a) What is the minimum height h_0 for which this shell will

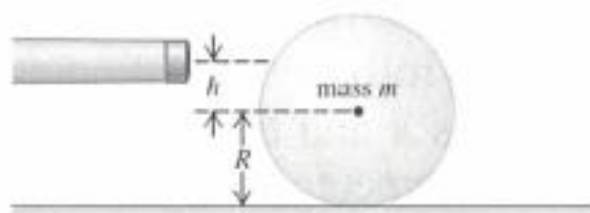
make a complete loop-the-loop on the circular part of the track? (b) How hard does the track push on the shell at point B , which is at the same level as the center of the circle? (c) Suppose that the track had no friction and the shell was released from the same height h_0 you found in part (a). Would it make a complete loop-the-loop? How do you know? (d) In part (c), how hard does the track push on the shell at point A , the top of the circle? How hard did it push on the shell in part (a)?



Ans: (a) $h_0 = 17/6 R$ (b) $11/5 mg$ (c) Yes (d) $2/3 mg$

Billiard Physics

A cue ball (a uniform solid sphere of mass m and radius R) is at rest on a level pool table. Using a pool cue, you give the ball a sharp, horizontal hit of magnitude F at a height h above the center of the ball (Fig. 10.37). The force of the hit is much greater than the friction force f that the table surface exerts on the ball. The hit lasts for a short time Δt . (a) For what value of h will the ball roll without slipping? (b) If you hit the ball dead center ($h = 0$), the ball will slide across the table for a while, but eventually it will roll without slipping. What will the speed of its center of mass be then?



Ans: (a) $2/5 R$ (b) $5F\Delta t/7m$

13. Rolling without slipping or sliding up the hill.

A ball with mass m , radius r and moment of inertia about its centre of mass βmr^2 is rolling along at speed v without slipping or sliding on a horizontal surface when it comes to a hill that rises at a constant angle above the horizontal. In which case will it go higher up the hill: if the hill has enough friction to prevent slipping/sliding or if the hill is perfectly smooth?

14. Rolling with slipping till without slipping

A uniform solid sphere with mass $M = 2.0$ kg and radius $R = 0.10$ m is set into rotation with angular speed $\omega_0 = 70$ rad/s. At $t = 0$ the sphere is dropped a short distance (without bouncing) onto a horizontal surface. There is friction between the sphere and the surface. Find

- (a) the angular speed of rotation when the sphere finally rolls without slipping
- (b) the amount of kinetic energy lost by the sphere
- (c) Show that the amount of kinetic energy lost in (b) is due to the work done by the frictional force that acts to cause the sphere to roll with slipping.

Answer: (a) 20 rad/s (b) 14 J

15. Rolling without slipping or sliding on a circular slope.

A small sphere of radius $r_0 = 1.5$ cm rolls without slipping on the track which is fixed on the ground as shown in Figure 3(a). The radius of the track is $R_0 = 26$ cm. The sphere is released from rest at height R_0 above the bottom of the track and rolls without slipping down the track.

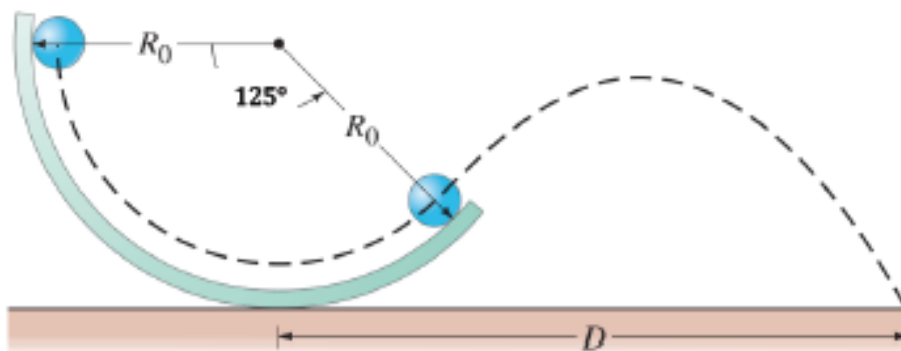


Figure 3(a)

- (a) What is the angular velocity of the sphere when it leaves the track after passing through an angle 125° as shown in Figure 3(a).
- (b) What is the net acceleration of the sphere the moment before it leaves the track?
- (c) At what distance D from the base of the track will the sphere hit the ground?

Answer: (a) 111.7 rad/s (b) 12.2 m/s^2 at an angle of 70.7° to the slope (c) 0.463 m